Student Attendance Management System

Course Name: Advanced Programming (OOPS)
Student Name: Ayush Debnath
Semester: Spring 2026

Problem Statement :

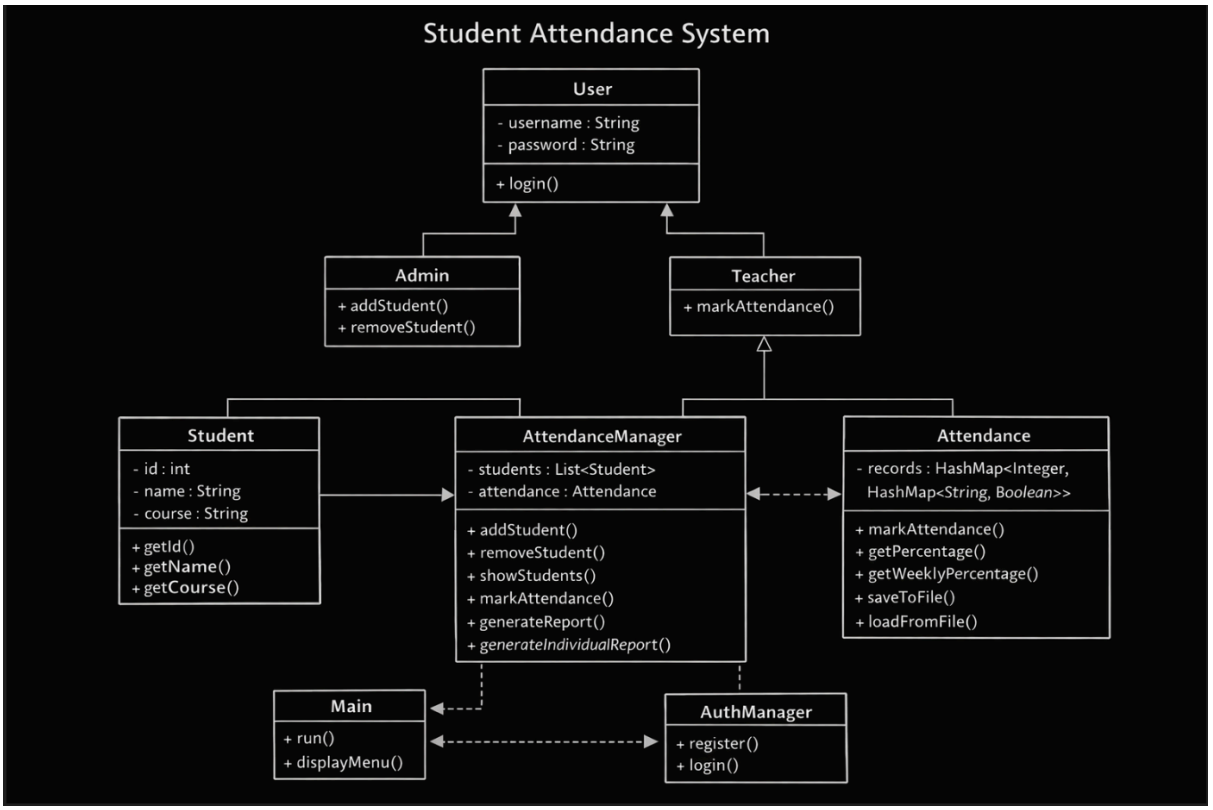
Attendance tracking is an essential part of academic institutions. However, many institutions still rely on manual methods or basic systems to record attendance, which leads to inefficiencies such as inaccurate records, difficulty in tracking attendance percentage, and lack of proper reporting.

Manual systems are prone to human errors, data inconsistency, and delays in processing. As the number of students increases, maintaining accurate attendance records becomes difficult.

To overcome these challenges, a **Student Attendance Management System** is developed using Java and Object-Oriented Programming concepts. This system automates attendance tracking, percentage calculation, and report generation while ensuring data accuracy and efficiency.

System Design :

UML Class Diagram



Description of Major Classes :

Student Class :

Represents a student with attributes such as ID, name, and course.Provides methods to retrieve student details.

Attendance Class:

Handles attendance records using HashMap.Stores student attendance day-wise and calculates:

- Overall percentage
- Weekly percentage

AttendanceManager Class:

Acts as the main controller of the system.Handles:

- Adding/removing students
- Marking attendance
- Generating reports

User Class (Base Class):

Defines common attributes:

- Username
- Password

Admin Class

Inherits from User.Can:

- Add student
- Remove student

Teacher Class

Inherits from User.
Can:

- Mark attendance

AuthManager Class

Handles authentication:

- User registration
- Login verification

Main Class

Entry point of the program.Provides menu-driven interface for user interaction.

Interaction Flow

1. User registers or logs in
2. System verifies credentials
3. Role is assigned (Admin/Teacher)
4. Admin manages students
5. Teacher marks attendance
6. System calculates percentages
7. Reports are generated
8. Data is saved to file

OOP Concepts Demonstrated :

Abstraction:

Used in: `User class` Defines common structure for Admin and Teacher

Inheritance:

Used in: `Admin`, `Teacher`Extends User class

Polymorphism:

Used in: login system / method overriding

Encapsulation:

Used in: `Student class` Private variables with getters

Implementation Highlights

Snippet 1: Attendance Logic:

```
records.putIfAbsent(studentId, new HashMap<>());
records.get(studentId).put(day, isPresent);
```

Explanation:
Stores attendance using nested HashMap for efficient tracking.

Snippet 2: Percentage Calculation:

```
return (present * 100.0) / total;
```

Explanation:
Calculates attendance percentage dynamically.

Snippet 3: File Handling:

```
BufferedWriter bw = new BufferedWriter(new FileWriter("students.txt"));
```

Explanation:
Used to store data permanently.

Snippet 4: Login System:

```
if(fileUser.equals(username) && filePass.equals(password))
```

Explanation:Validates user credentials from file.

Use of Collections

- HashMap → Attendance storage
- ArrayList → Student list

Testing & Error Handling

Test Cases

- Add student
- Mark attendance
- Generate report
- Login with valid/invalid credentials

Edge Cases

- No students
- Duplicate student ID
- Invalid login

Error Handling

- Input validation
- File existence checks
- Safe parsing

Git Workflow

- Initial setup
- Added classes
- Implemented attendance logic
- Added login system
- Added file handling

Conclusion & Future Scope

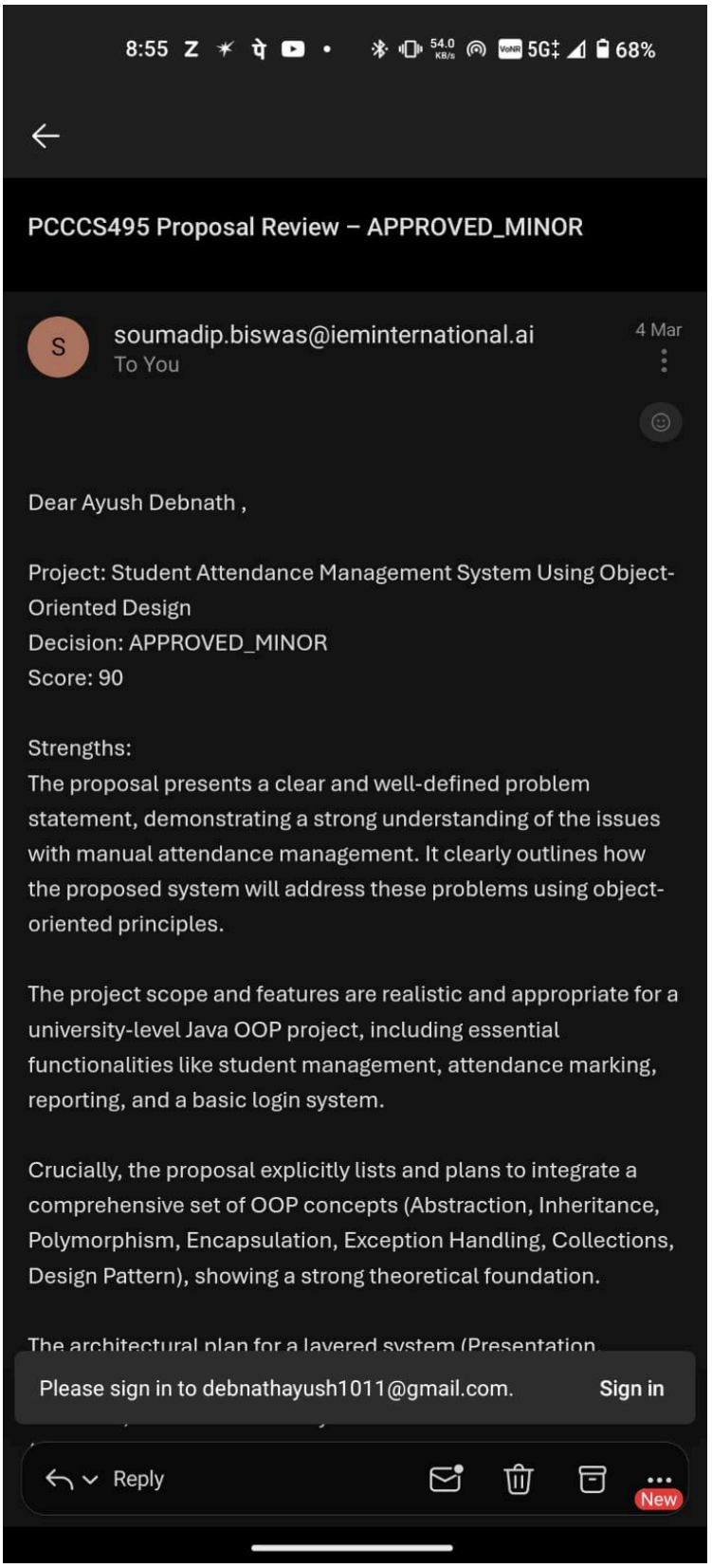
Conclusion

The Student Attendance Management System successfully demonstrates the use of Object-Oriented Programming concepts using Java. It provides an efficient and reliable way to manage attendance records and generate reports.

Future Scope

- GUI implementation (Swing/JavaFX)
- Database integration (MySQL)
- Password encryption
- Mobile/Web version
- Export reports (PDF/Excel)

Appendix:



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PCCCS495 Proposal Review – APPROVED_MINOR

S soumadip.biswas@ieminternational.ai 4 Mar
To You

Dear Ayush Debnath ,

Project: Student Attendance Management System Using Object-Oriented Design
Decision: APPROVED_MINOR
Score: 90

Strengths:
The proposal presents a clear and well-defined problem statement, demonstrating a strong understanding of the issues with manual attendance management. It clearly outlines how the proposed system will address these problems using object-oriented principles.

The project scope and features are realistic and appropriate for a university-level Java OOP project, including essential functionalities like student management, attendance marking, reporting, and a basic login system.

Crucially, the proposal explicitly lists and plans to integrate a comprehensive set of OOP concepts (Abstraction, Inheritance, Polymorphism, Encapsulation, Exception Handling, Collections, Design Pattern), showing a strong theoretical foundation.

The architectural plan for a layered system (Presentation

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